Phishing Awareness

Training Module: Recognition, Tactics, and Best Practices

CodeAlpha Internship Task 2

The Anatomy of Phishing

Understanding the threat landscape and why human awareness is the best line of defense.

What is Phishing?

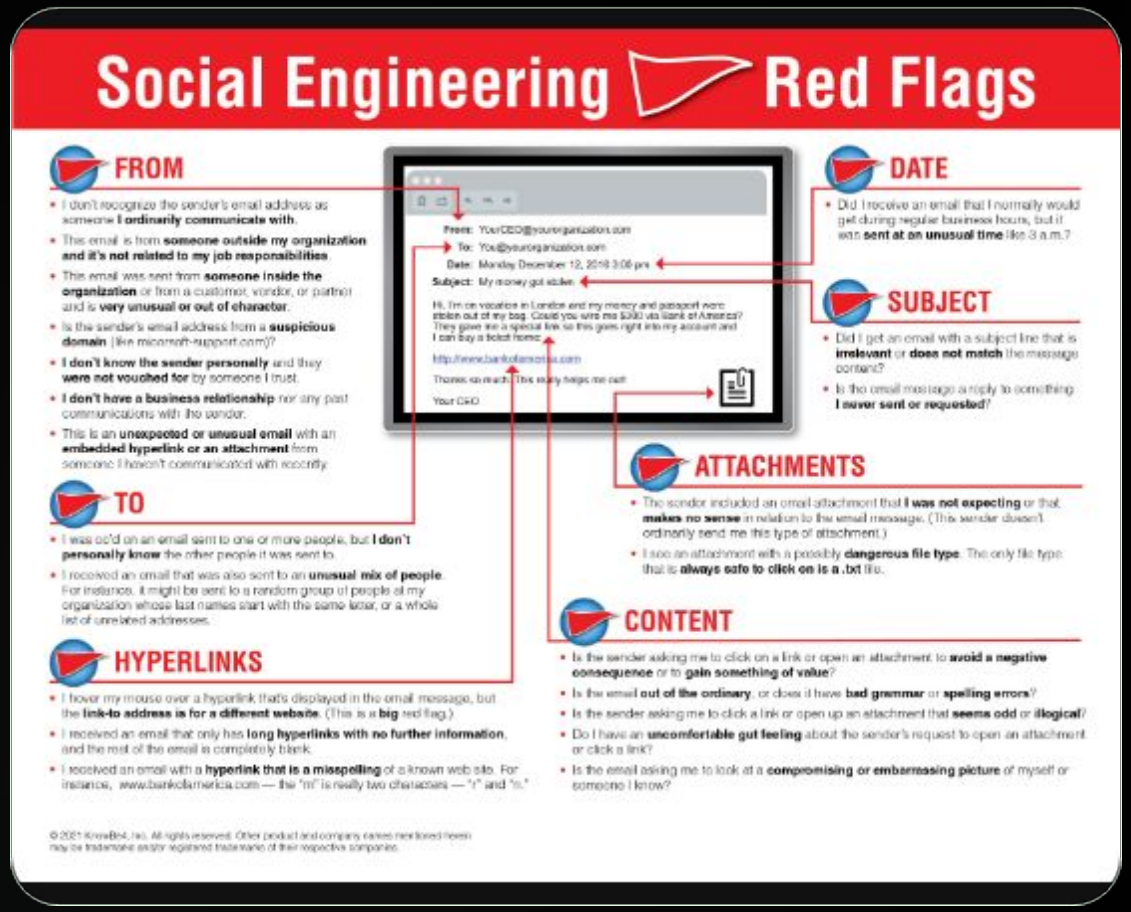
Social Engineering

A type of cyberattack where attackers deceive users into revealing sensitive information by pretending to be a trustworthy entity.

Primary Methods

Commonly delivered via email, text messages (Smishing), or phone calls (Vishing) to harvest credentials or install malware.

Spotting the Red Flags



- **Mismatched Sender:** Look closely at the domain (e.g., micr0soft.com).
- **Extreme Urgency:** Language like "Immediate action required" or "Account suspended."
- **Generic Greetings:** Using "Dear Customer" instead of your actual name.
- **Suspicious Links:** Hover before you click to see the true destination URL.

Psychology of the Attack

Authority

Impersonating a CEO or IT manager to demand compliance without question.

Scarcity

Creating fear that an opportunity or access will be lost if you don't act now.

Familiarity

Leveraging public data to build false trust through spear-phishing.

The Massive Scale

90%

Of All Data Breaches

Phishing is the #1 entry point for ransomware and large-scale corporate data theft.

- 3.4 Billion phishing emails sent daily.
- Average cost of a breach: $4.45 Million.
- Human error remains the weakest link.

Beyond the Inbox

Spear Phishing

Highly targeted attacks using specific details about the victim.

Smishing

SMS phishing messages often pretending to be delivery services or banks.

Quishing

Malicious QR codes on posters or mailers that steal credentials.

Real-World Incidents

Victim Organization	Attack Vector	Documented Impact
Change Healthcare (2024)	Credential Phishing	100M+ Users Exposed
Pepco Group (2024)	BEC / Whaling	€15.5 Million Stolen
MGM Resorts (2023)	Vishing (Phone Scam)	$100M+ Operational Loss

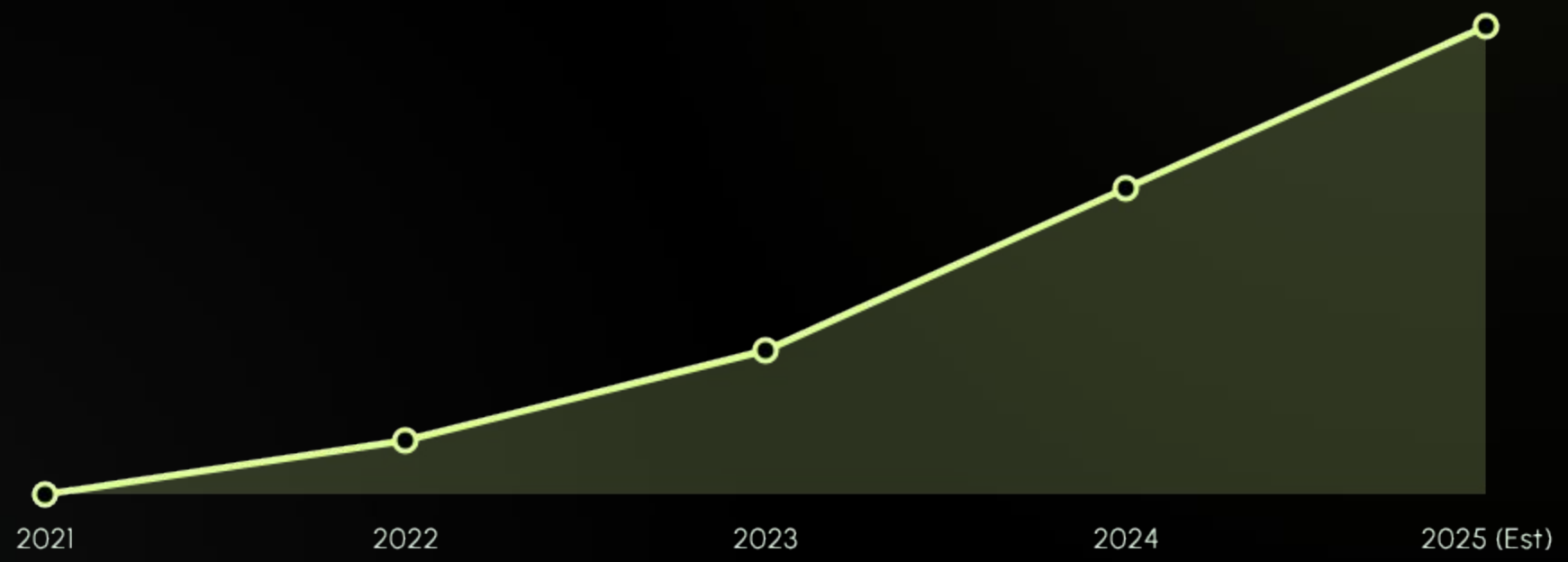
Identifying Fake Websites

- **URL Discrepancies:** Look for "amazom.com" or "g00gle.net".
- **The Padlock Myth:** HTTPS only means the connection is encrypted, not that the site is legitimate.
- **Poor Visuals:** Broken links, blurry logos, and inconsistent branding.
- **Unusual Pop-ups:** Immediate requests for login or card info.

⚠️Security Pro Tip:

Never navigate to a login page via an email link. Always type the official domain directly into your browser address bar.

Phishing Volume Trends



Explosive growth in phishing volume (2021–2025) driven by AI automation.

Interactive Knowledge Check

Q: You receive an urgent email from your "CEO" asking you to purchase gift cards for a client immediately. What is the safest action?

A) Buy them and send the codes to the email address provided.

B) Reply to the email asking for more details.

C) Call the CEO or verify through an official office channel.

Your Defense Roadmap

- Enable **Multi-Factor Authentication (MFA)**.
- Never share passwords or sensitive info via email.
- Always hover over links to verify destinations.
- Report suspicious activity to IT immediately.

Questions?

Stay vigilant and protect the network.

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